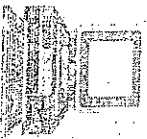


1. **HYBRID COMPUTER:** This type of computer is that, whose mode of processing combines both analogue and digital, that is, mechanical and electrical aspects.

Examples are Desktop computer, laptop computer, etc.



Desktop computer

TYPES OF HYBRID COMPUTER

Hybrid computer is classified into three basic classes as

- a. The Macro Hybrid computer
- b. The Mini Hybrid computer &
- c. The Micro Hybrid computer.

The Macro Hybrid computer

The macro hybrid computer, owing from the name "Macro" which means *big*, is the biggest of all the three hybrid types. It has got many advantages over the others. It has the following characteristics:

1. largest size
2. fast processing speed
3. great storage capacity
4. support for multiple users.

Example of such computer is the Mainframe computer. They are found in Stadium (as scoreboard) and in meteorological department of aviation used in the presentation of weather forecast.

The Mini Hybrid computer

This computer is a medium size between the macro and the micro. They have the following characteristics:

1. largest size
2. fast processing speed
3. great storage capacity
4. support for multiple users.

The Micro Hybrid computer

This computer is the smallest of all. It has the best portability among the others and possesses the advantage of large memory capacity over all the hybrids. It is also classified into various sizes, capacities as well as speeds of velocity. This classifications in models are:

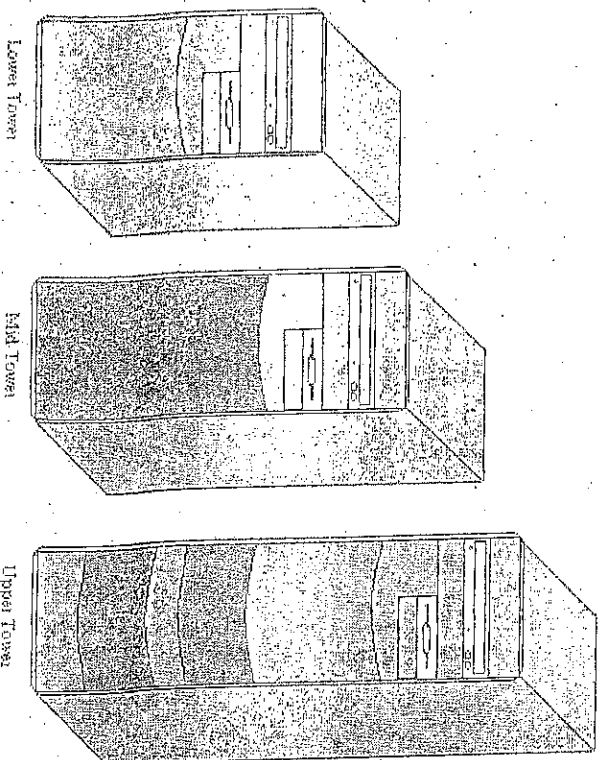
- ☐ the desktop model
- ☐ the laptop model

- the notebook model &
- the palmtop model.

TYPES OF MICRO COMPUTER BASE ON CASE TYPES

A first consideration is the environment for using the system. The options are:

TOWER --- This type has a large, vertical case with plenty of space inside for the inclusion of extra equipment. The case should have extra drive bays, so that CDROMs, extra hard disc, etc can that extra equipment be easily fitted. The main board used in these models should have plenty of unused



expansion slots, so that extra equipment cards can be plugged in. This is the first choice for a system that carries out a lot of the office's extra activities. Examples of this would be using the machine as the server in a local area network or using the machine as the offices main resource for fax cards, modems, scanners, tape backup, e.t.c. The unit normally sits on the floor and the monitor sits on the desktop. Now supplied in AT and ATX forms.

Mini-Tower

This type has a vertical construction as above but is not so tall. It can store less than a tower but is more expandable than the average desktop machine. It is a good compromise for size and expandabilities. It usually sits on the desktop with the monitor close by.

Desktop --- This is the most common of all case types and is a horizontal box of varying dimensions, on to which most user place the monitor.

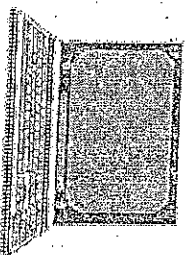


Desktop

Low-Profile Desktop. This is smaller 'slim line' version of the above, specially miniaturised to occupy less desk space; they are often also of the 'small footprint' variety, ie they have smaller breadth and depth dimensions than ordinary desktop machines and therefore takes up less of the desk area. They are neat machines and pleasing to the eye but they have a major drawback with their expandability. Since the case is of a low profile, there is not enough height in the machines to plug any expansion cards straight into the board's expansion slots. A special 'plane' boards has to be fitted vertically into one of the expansion slots. The expansion cards then plug horizontally into one of the expansion slots on the plane board. This normally results in fewer slots being available in low-profile models and, in some cases, the power supply has to be unscrewed to allow the extra cards to be fitted. Unless future expansion of the machines is ruled out, this would not be the first choice for office machines.

Portable - These machines are about the size of a portable typewriter and are designed for use with internal batteries, the car's 12 volt lighter socket or a mains adaptor. This means that they are very portable and are in regular use by those who require computing facilities away from the office. They have small screens built in to the case lid and these are inferior to standard monitors, particularly for prolonged use.

NoteBook - A notebook is an even smaller version of the portable and is around the size of an A4 sheet of paper. Unlike the portable, they are small enough to carry in a brief case along with other materia and are now more popular than the portables.



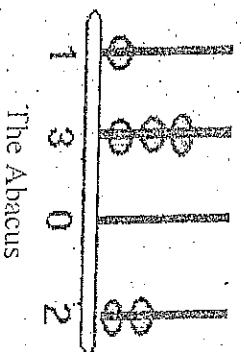
A Notebook

THE HISTORY OF COMPUTERS

Before now efforts have been made several years to develop calculating machines. These machines were made to achieve goals by various scientists, mostly mathematicians who contributed immensely in computer and mathematical researches. These devices earlier produced or discovered were

- (1). The Abacus
- (2). The Pascaline
- (3). The Stepped Reckoner
- (4). The Analytical Engine
- (5). The Tabulating Machine

1. **ABACUS**:- It is the first non-mechanical calculating device which consists of beads strung on wire or wooden rods. Using an Abacus, one could quickly and accurately add, subtract, multiply and divide large numbers. Later people could no longer use the Abacus for all forms of calculations, so the need for other devices became necessary.



The Abacus

2. **PASCALINE**:- It is the first mechanical calculating machine. It was invented in 1642 by a French mathematician named "BLAISE PASCAL". It consists of a combination of interlocking wheels. The wheels contain printed numbers 0 to 9. When the first wheel completely rotates from 0 past 9, it automatically causes the wheel to advance to the next and when the second wheel moves past 9, it costs the third wheel to turn and so on. It could only add and subtract numbers. The difference between the Abacus and the Pascaline was that, Abacus could add, subtract, multiply and divide numbers, while Pascaline could only add and subtract numbers.



A Pascaline

3. **STEPPED RECKONER**:- In addition to the Pascal's adding machine (Pascaline) a German mathematician "GOTTFRIED WILHEM VON LEIBNITZ" in 1694 invented a Stepped Reckoner. The machine used cylinders marked with numbers and contained 9 rows of pegs or teeth, the first row had 1 peg, the second had 2 pegs, the third had 3 pegs and so on, up to 9, the machine was more complex than the Pascaline and could do more. The Pascaline could only add and subtract but, Leibnitz's machine could multiply divided and find square root of numbers.

4. ANALYTICAL ENGINE--

An English mathematician called "CHARLES BABBAGE" in 1822 attempted to design a device that would capably be of performing complex calculations, such as finding power of given numbers. This machine was to be called the Difference Engine. Later he got another idea of inventing a machine which he called the "*Analytical Engine*" it could calculate numbers, store data and print solution to problems

The Babbage's machine was to have 3 parts

- 1 the store
- 2 the mill and
- 3 the control

THE STORE:- This would hold data needed for computation

THE MILL:- This would process data

THE CONTROL:- This would be the automatic operator

Unfortunately, babbage died without actualizing his plans of inventing the machine which would operate independently of human operator, yet his proposed invention was an important one. The machine Babbage planned to build was to be a steam-powered mechanical computer and the programs that would control it were coded on punched cards.

PUNCH CARDS:- This card was designed by JOSEPH JACQUE.

5. TABULATING MACHINE--

In 1880, an employee of the United States Census Bureau, "HERMAN HOLLERITH" developed a machine to help tabulate census data. The machine used punched cards and electricity. The data from the census was first punched into cards and these were then processed by a tabulating machine. He also developed a machine to sort these information. As a result, the Bureau processed the 1880 census data in less than 3 years as against the projected 9 years. Hollerith later left the census bureau to form the Tabulating Machine Company. This company later became the *International Business Machine cooperation* (IBM).

THE GENERATION OF COMPUTERS

In the first-half of the twentieth century, computers which could perform better than the mechanical calculators evolved. They were electronic machine capable of solving complex mathematical problems. These computers had two key features that the early calculating machine did not have. These vital features were:

- a. ability to be programmed to carry out a sequence of calculations or to perform several tasks, and also the
- b. ability to store information in memory.

First Generation Computers

The computer called MARK I was invented at the Harvard University in 1944 by Professor Howard Aiken. The MARK I was originally called the Automatic Sequence Controlled Calculator. It was the first digital computer. The machine accepted input coded on punched cards and was not fully electronic in operation but could carry out calculations in tremendous speed. It used signal in combination with mechanical gears, switches and wheels. The MARK I also had a great storage capacity. Nevertheless, it became obsolete a few years later. Another invention paid and built for the United State Army was the ENIAC (Electronic Numerical Integrator and Computer) it was built by a team of engineers led by John Mauchly and J. Presper Eckert in 1946. This invention marked the beginning of the first generation of electronic computer. This generation was characterised by the following:

- The use of vacuum tubes as the major switching element
- Very large size for quite small capacity
- The vacuum tubes used large amounts of power, thereby generating a tremendous amount of heat.
- The construction of the first commercially used computer called UNIVAC (Universal Automatic Computer)
- Programming in machine language

Second Generation Computers

The second generation of electronic computer came up in 1959. The greatest development that followed this generation was from a transistor developed to service as the major switching element in lieu of the vacuum tube in the first generation. The transistors are much smaller and operate more efficiently than the vacuum tube of 1st the generation.

The characteristics of the 2nd generation computer were:

- a. Computer were smaller faster and less expensive.
- b. It witnessed a big leap in the use of computers.
- c. Efforts were made to develop high level programming language like FORTRAN, COBOL.

Third Generation Computers

The third generation of computers started in the year 1964 and was brought about by the development of the integrated circuit (IC). It resulted in great miniaturization of computers and further increase in the speed, efficiency and capabilities of computers. This generation was characterized by the following.

- Much smaller and faster computers
- The development of minicomputers beginning with DEC.PDP-8
- Great advances in software development
- Increase in areas of computer application
- Development of additional programming languages like BASIC
- The development of operating systems.

Fourth Generation Computers

The fourth generation of computers started in 1970 till now. This was brought about by a technology called Large Scale Integration (LSI) on integrated circuits, which enabled greater functions to be achieved on computers at faster speed and smaller size. This generation has the following characteristics

- The introduction of the microprocessor by the intel corporation in 1971.
- The introduction of the first microcomputer.
- Rapid development and expansion in the use of computers.

The fifth generation are presently under development. This generation would have a different structure from the other ones. Its greatest characteristics is of the fact that it will enable computers to process "intelligence" and able to "think" like human.

HARDWARE

A hardware can be defined as the structural or physical materials of a computer. In other words, it is the frame-work of a computer. Hardware is classified into three units based on their functions in a computer. These unit are:

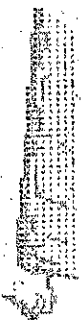
- Input unit device
- Processing unit device
- Output unit device

INPUT DEVICE

This device enhances information needed for processing to be put into the computer through keying, clicking, dragging, scanning and other ways. Example of such devices are:

1. keyboard
2. mouse
3. joystick
4. light pen
5. scanner
6. digital camera

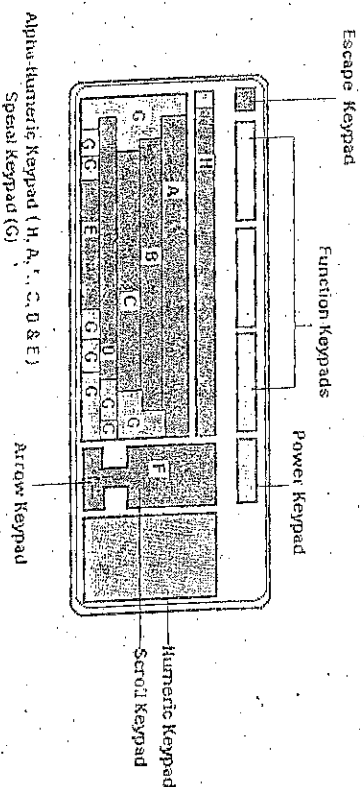
A. Keyboard



A Keyboard

The keyboard is an input device that enables or allows the keying of information into the computer for processing. A keyboard comprises buttons or keypads of specific and different functions from each other. The keypads are basically classified into various divisions based on their mode of functions and arrangements. These classification are:

- **Functional Keypads**—These are keypads found on the uppermost top alignment of a keyboard. They are F1, F2, F3-----F12. They serve various functions. They could be used to run (eg. F5), and compile a program software as required by the programmer.
- **Numeric Keypads**—This could be called a calculator keypad due to its keypad arrangement as in a calculator. It comprises the arithmetic signs(+ - x ÷ = < > etc) in addition to numbers from 1-0.



• The

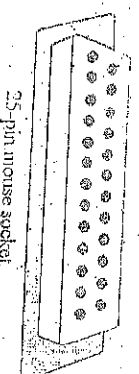
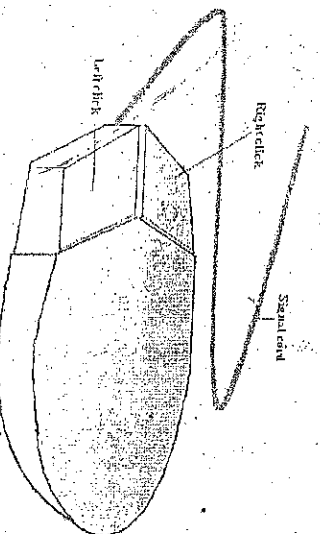
Alpha-Numeric Keypads- This is called "typewriter keypad" it combines both numbers and alphabets. Other than the numbers are the space-bar, backspace, punctuation signs (., / ?) etc.

- **Special Keypads**-These are used to move cursor and pages. These are page up, page down home, end, insert, pause, print, scroll etc. enclosed in the scroll keypads are the directional keys or arrow keys, or mouse keypad

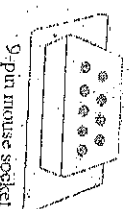
- **Power Keypads**- These control the power system of the computer, that is, On and Off. The keys are inscribed with sleep, wakeup, and power.

4 Mouse

The mouse is an input device that enhances the passage of data or information in the form of signal into the computer system by the way of *clicking* and *dragging*. The mouse helps in the running and control of various programs. The mouse may be of the PS/2 or the Serial type in terms of its plug. For the serial socket, it usually has 9 pins and is marked as 'Serial', 'COM1', or 'RS232'. Sometimes the socket is already used (eg a Modem) and the mouse connects to the second serial socket which has 25 pins via the 25-pin adapter.



25-pin mouse socket



9-pin mouse socket

PROCESSING DEVICE

The processing device comprises the CPU which, contains many components that work hand-in-hand to ensure efficiency of processing, thus, there is no specific device other than CPU. In the CPU, there are other devices that ensure the storage of data or information after processing. These memory or storage devices are classified into:

- Primary memory devices
- Secondary memory devices

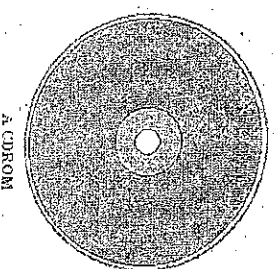
Primary Memory Devices

These devices hold on information temporarily in a computer once the power is on but, could be lost when the power is off. Some of them store hard-wired information

produced by the manufacturer and the information stored in them could not be altered. Example of such hardwares are:

RAM---The Random Access Memory stores information temporarily. The device must be installed in a system to make it work. It has a *volatile function*, that is, the data in it could be retrieved when the power of a computer system is On but, lost when the power is Off. RAM are produced in various capacities ranging from 8MB to 256MB, and above with its availability in *SIMM* (Single Interface Memory Module) and *DIMM* (Double Interface Memory Module).

ROM---The Read Only Memory contains unalterable data. The ROM only allows stored information to be read. It is grouped into PROM (Programmable Read only Memory) and EPROM (Erasable Programmable Read only Memory). The former stores hard-wired information, which remains permanent as designed by the manufacturer as default settings without any provision for alteration, while the latter stores information that could be changed to the suite or desire of the user.

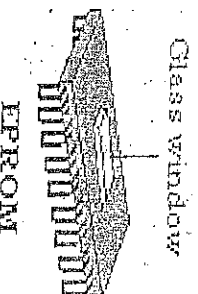


A CDROM

Secondary Memory Device

This device is, in another word, known as a back-up device. It allows both reading and writing of information. Information are stored permanently unlike the "RAM" except when the device is damaged. Example of such devices are:

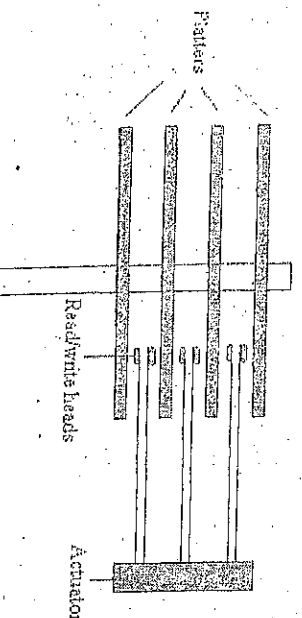
- Hard disc
- Floppy diskette
- CDROM



Hard Discs

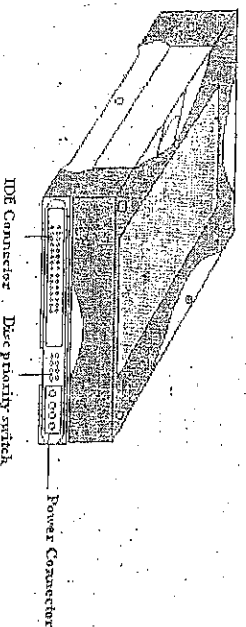
Hard discs systems are composed of :

- A sealed drive unit
- A disc controller electronics, either on a controller card or built into the drive.
- Connecting cables between the drive and the computer motherboard.



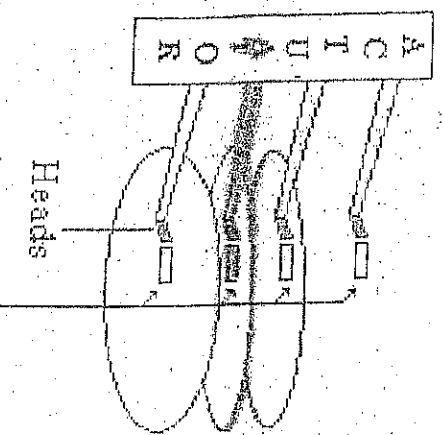
The hard disc units are comprised of two or more discs in the one sealed unit, while the floppy is a single plastic disc. The extra discs boost the drive's storage capacity and each of these discs is known as

a 'platter'. The term applies to both sides of that particular disc. So, if a drive had three platters, it would contain 6 sides. Like floppies, area of the hard disc's coating are magnetised and demagnetised to store the data. The disc is aluminium and coated with ferric oxide or cobalt oxide, but in some newer models, the disc is made of glass. Modern drives can have many platters, depending on the capacity of the drive. A 210MB hard disc may have six platters (i.e. 12 sides), while Fujitsu's 8.8GB model has ten platters



IDE Connector Disc priority switch Power Connector
A Hard Disc Drive

Cylinders Many specifications and publications refer to the term "cylinders" when describing a disc's construction. Mostly, the term is used freely as the equivalent of 'tracks'. With hard discs there are a number of different platters and the heads are mechanically linked and are therefore placed over a set of track at any one time. The



Cylinder

a park command, or newer machines are powered down, the read/write heads are moved to sit in the landing zone so that they will not be able to scratch the surface of any sectors used for storing data. A drive rated as 771 tracks might only format the disc to 770 tracks, using track 771 as the landing zone. Some systems also use spare inner tracks to make up bad tracks. This can result in different utilities displaying different results: some are counting the tracks for data while others are counting the total tracks.

With IDE and SCSI systems, the electronics can be used to hide the real surface structure from the machine. As far as the machine is concerned, the drive has say 953 tracks, with each track containing 36 sectors. This will indeed be the 'logical' view given to the machine by the drive. In fact, the drive may well have a completely

collection of tracks covered is known as the driver's 'cylinder'. In a hard disc, each platter will have an identical set of cylinder numbers and these number from zero upwards. So cylinder 2 would be the third track on side 0, the third track on side 1, the third track on side 2 and so on. ~~When a floppy disc has only two sides, the term is never used.~~ The inner track are smaller than the outer tracks, so the data stored more densely in the sectors of these tracks. The density of the innermost track determines the drive's maximum capacity. When a disc is constructed, it is provided with one or more extra inner tracks. These are used to provide a landing zone for the read/write heads. When older machines are given

different physical layout. Since the disc's outer tracks cover a much greater surface than the inner tracks, the outer tracks can be used to store up to three times more sectors per track than the inner tracks. This is known as 'zone bit recording' and provides greater storage with improved reliability. The exact internal layout need not be known by the machine or by the user, since the electronics on the drive make the drive appear to have the normal regular track and sector layout.

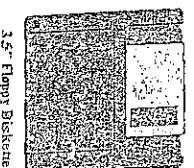
Heads

Each side requires its own read/write head and these number from zero upwards. All the head are mounted on the same actuator arm, so that they all move in unison as a single unit. When one head is on the sixth track in from the edge, then all the other heads will be on the sixth track of their disc sides. This is used to achieve economy of head motion. All the sectors in the same cylinder number (of all platters) will be used, before moving the head inward to the next cylinder number.

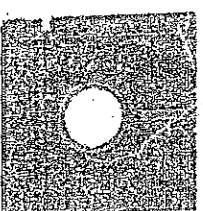
The maximum read without moving the head to another track is calculated by multiplying the number of sides by the number of sectors in the track, this is then multiplied by the sector size, usually 512 bytes.

Floppy Discs

Floppy discs for PCs come in two sizes- the older 5.25" and the current 3.5" size. This refers to the diameter of the discs inside the protective casing. Both discs are made from mylar plastic coated with ferric oxide. The floppy drive, is normally as the A: drive; if a second floppy is fitted in a machine, it will be known as the B: drive. The hard disc is called the C: drive.



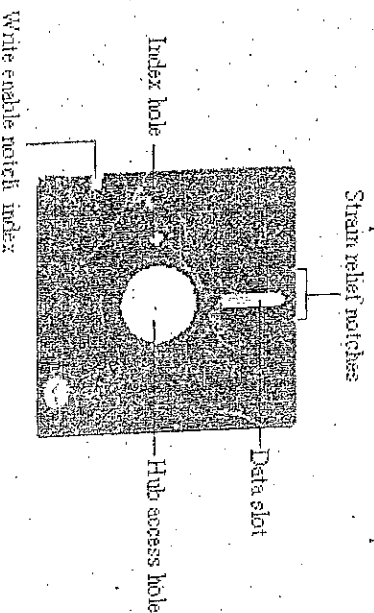
3.5" Floppy Diskette



5.25" Floppy Diskette

5.25" Discs

The disc sits in a vinyl wallet, or jacket, or protect the magnetised surface from exposure or grease. The disc spins at 300rpm inside the jacket, so the jacket lining is coated with a soft, non-woven material that is very lightly lubricated to minimise friction. The write-protect notch allows or prevents writing to the disc. A tab can be

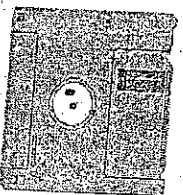


placed over the notch to disable any writing to the disc. In this way, important data can be written to the disc and then protected against accidental erasure or modification. When the tab is removed, the disc is again able to be written to. The wallet has a slot in both sides, to expose the surface so that the

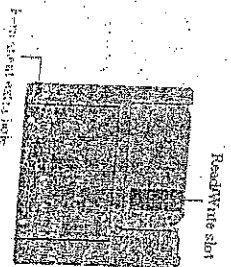
read/write heads can reach the magnetised coating on the disc. The read/write head is brought close to the disc surface by the action of closing the disc drive lever after the disc is inserted. When not in use, the disc should be kept inside its cardboard sleeve to protect the exposed area. This disc also has an 'Index Hole'. This is a hole that is cut both in the jacket and in the disc itself. As the disc rotate inside the jacket, a point will be reached when both holes will align. At this point, a light from one side of the disc is able pass through the hole and be detected by a light sensor on the other side of the disc. The timing of these pulses of the light tell the computer exactly how much rotation has taken place. This information, when combined with the head movement, can place the read/write head on any sector on any track with great accuracy. In the centre of the disc there is a 'Hub Access Hole'. This hole allows the mechanism's drive hub to hold the discs in the set of 'Strain Relief Notches' on the disc wallet. These are cut in the jacket close the head access slot, to reduce creasing when the jacket is stretched. The lower density product i.e. Double Density uses 40 tracks and 9 sectors per track. This means that there are 360 sectors on each side, giving a total of 720 sectors on the entire disc. Since each sector is 512 bytes, the disc can store 360k of data. The higher density version described as Quad Density increases the number of tracks to 80 and the number of sectors in each track to 15 giving a storage capacity of $80 \times 15 \times 2 \text{ sides} \times 512 \text{ bytes} = 1.2 \text{ Mb}$. A double density disc has a coercivity of 300 Oersted while a quad density disc has a coercivity of 600 Oersted.

When data is written to a disc, the level is taken as being 100%. When that same data is read from the disc, its level is lower than the original figure this reduction is known as 'clipping level'. A known brand disc normally returns a clipping level of around 50% while bulk disc are around 40%. Disc drive, however, are able to adequately cope with clipping levels down to about 20%.

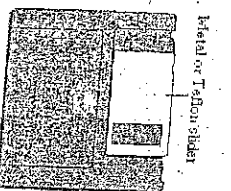
The inner disc of this type of floppy is constructed simply to the larger floppy, except that it has smaller dimensions and a higher 'Quallit coating'. Its can store more than the 5.25" despite its smaller proportions. The 3.5" is more robust since it is housed inside a rigid plastic case. they read/write slot is protected by a metal



Write-protect hole
BACK VIEW



Read/Write slot
FRONT VIEW

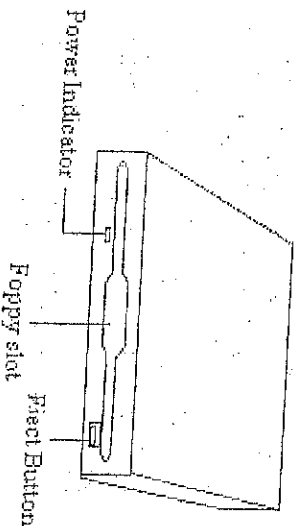


teflon slider that cover the slot when the disc is outside the drive, when the disc is placed in the drive, the drive mechanism push the slider back so that read/write head can gain access to the magnetic surface. 3.5" discs have a write-protect hole that can be covered or uncovered by sliding a plastic cover over it. When the hole is covered, the disc can be written to. This is the opposite of the 5.25" disc that can only be written to when its write-protect notch is uncovered. Double Density discs have 80 tracks with 9

80 x 9 x 2 sides x 512 bytes = 720k. High Density disc have 80 tracks and 18 sectors in every track. They have double the capacity of double density disc giving a capacity of 1.44Mb. Some dealers offer '2Mb disc'. These are just high density disc, being quoted with their unformatted capacities; when the formatting data is placed on the disc, the disc is left with a usable capacity of 1.44Mb. The coercivity of a 720k disc is 600 Oersted and a 1.4Mb disc coercivity is 700 Oersteds.

A high density disc has a permanent extra hole punched in its body; this hole is not designed to be covered and uncovered like the write protect hole. The hole's presence or absence indicates whether the disc is a double density or high density version. The drive decides to use high or low head currents by checking for the presence or absence of this hole.

Floppy Disc Drives



A 3.5" Floppy Drive

A floppy disc drive is a device that rotates the floppy disc through several revolutions along the disc's track to enhance its reading and writing functions. The slotted floppy disc is driven round by the drive in a spinning process at three hundred ~~revolutions per minute (300rpm) inside the its jacket,~~

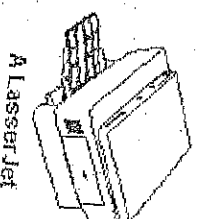
OUTPUT DEVICE

This device enables the outcome of the processed data to be seen or heard. Example of such device are

1. Monitor
 2. Print
 3. Plotter
 4. Multimedia speaker/head piece/mouthpiece & microphone e.t.c.
- In the output of a processed data, the unprinted copy, i.e the data seen on the screen is called **Softcopy** while the printed is called **Hard copy**

Printers

This an output device that enables data or information to be put or written on papers, that is, it enhances hardcopy production, printer is classified into;



A Laser Jet

- Dot-matrix-which uses inked ribbon
- Deskjet -which uses ink in a plastic case
- Laserjet- which uses powdered carbon in a toner
- Daisy wheel & thimble which uses inked ribbon.

Printer are classified into:

Impact—Use physical impact to transfer character to the paper using inked ribbon.
Non-impact--- Do not use physical impact to transfer character to the paper and there is no inked ribbon. Example of such are deskjet and laserjet while daisy-wheel and thimble and Dot Matrix are impact printers.

Characteristic Of Impact Printers

- It prints characters per character (12-14) per sec
- It makes physical noise
- It is slower in speed of Printing
- It uses inked ribbon
- It is economical

Characteristic Of Non-Impact Printer

- It prints pages per minute
 - It makes no physical noise
 - It is faster in printing
 - It uses inked (deskjet) and toner (laserjet)
 - Laserjet is economical while deskjet is expensive
- It is certain that the deskjet has the advantage of printing color output over the laserjet.

Monitors

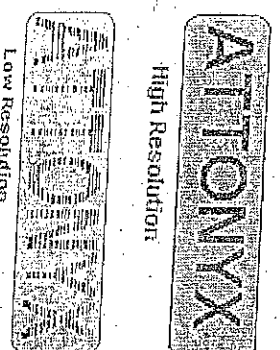
This is a primary output device that enables already processed data or information to be displayed. It is classified based on colour mode into;

- Monochrome
- Synchrone

The classification of monitors are based on the quality known as resolution or pixel.

A PIXEL

A pixel or pixel are lines horizontally across or vertically down the screen. For instance, a uniform screen display of 4:3 must be maintained at the same rate to yield good quality graphics. This means that 4 pixel across and 3 pixels down the screen must be attained to have a good quality of pictures or images. The number of pixel give



the phenomenon of a resolution. A pixel is the unit of or measurement of a resolution. Thus, resolution is entirely different from colour mode. Monochrome and synchrochrome are classification based on colour mode.

MONOCHROME

A *chrome* means "colour" while *Mono* means "one". Thus, monitors have one colour mode and one background display. Example of such

- MDA (Mono Display Adapter) - 80x25 characters
- Hercules—720x350 pixels.



A Monitor

SYNCHROME

This monitor supports more than one colour mode background display are:

- CGA (Colour Graphic Adapter) 2-4 colour mode and 640x200 to 320x200 pixels respectively.
- EGA (Enhanced Graphics Adapter) 16 colour mode and 640x350 pixels.
- VGA (Video Graphics Adapter) - 256 colour mode and 640x480.
- SVGA (Super Video Graphics Adapter) above 256 colour mode and 800x600 pixel or 1024x768 pixels
- XGA (Extended Graphics Adapter) 256 colour mode and 1024x768 pixels
- EVGA (Enhanced VGA) - 768 pixels.

SOFTWARE

A software can be defined as a series of coded instructions that directs the computer on what to do. A software is like a soul to the computer and without it, a computer is deemed functionless, and unimportant. It is the software that coordinates the entire activities of the hardware of a computer system. It is classified into various kinds based on their application purposes. A program is a sequential instruction or coded language which is written and compiled to communicate a computer to its hardware. When these programs are differently written and collected, they are compiled into software. So a **software** is the collection of various programs. The act of writing a program is called **programming** and the writer of programs is called a **programmer**.

TYPES OF SOFTWARE

Software are classified into:

- Operating system software
- Program packaged software
- Application packaged software

Operating System Software

This software coordinates the working ability of the computer. It enables the booting and configuration of a computer. Example of operating system software are;

- Dos (Disk Operating System)
- Windows 95, 98, 2000, ME, and XP
- Linux
- /cnix
- Nobel
- Unix

Program Packaged Software

This software enables the writing and development of other programs or possibly software. With program packaged software, a programmer could design various software particularly, application packaged types of software. Examples are;

- BASIC (Beginners All-purpose Symbolic Instruction Code)
- Dbase (Data base)
- QBASIC (Quick Basic)
- Visual Basic
- Visual Dbasic
- FORTRAN (Formular Translation)
- COBOL (Common Business Oriented Language)
- FOXPRO
- Boland c ++
- Pascal
- Oracle
- Turbo c ++

Application Packaged Software

This software performs a specific operation as designed to do. With various application packages, medical doctor, engineers, architects etc could develop so many things in their professions.

Application software is classified into

- Word Processing
- Spreadsheet
- Database Management system
- Utility.

❖ Word Processing

In a word processing software, a computer user could type or edit letters or sentences. Among the editing includes typesetting such as underlining, bolding, italicizing, increasing/reducing, bulleting, e.t.c. of alphabet which could be enhanced by highlighten the word needed. One could also draw a table of many columns and rows in order to insert numbers or letters. Examples of such software which are commonly found in almost all the windows- base computers are

1. Microsoft word
2. Corel word perfect
3. Microsoft power point
4. Microsoft paint
5. Corel draw
6. Adobe page maker
7. Microsoft notepad/wordpad, etc.

❖ Spread Sheet Application Software

It is a software that deals with all kinds of calculations. Other arithmetical and statistical calculations could be done such as completing the table of a quadratic or linear function as well plotting its graph. One could create a histogram, pie-chart or bar-chart for a given table. Example of such spreadsheet are

1. Lotus 1,2,3
2. Microsoft excel

❖ Data Base Management System.

It helps as a storage medium which include the inputation of records or examination results and personal resume. Example of such software includes:

1. Microsoft access.

❖ Utility Application Software

This software enables the working proper of a computer system by a way of configuring or customizing its abnormalities. E.g of such utility software are:

1. Anti-virus
2. Drivers
3. Diagnostic
4. Disk Manager

Anti-virus

---it's a special software designed to detect and remove a virus (software designed to attack files and operating systems).

Driver

---It is a software that communicates a computer with an attached or connected hardware such as printer, scanners e.t.c.

PROGRAMMING LANGUAGE

Basically, programming language is classified into high level language and low level language. Before we go into detail explanation of these languages, the computer numbering system would have to be discussed elaborately.

The Numbering System

It is very important to have the concept of how a computer system works, especially in its principle function of the binary number representation. It also uses a variety of formats in its storing of data. Integer numbers are stored differently, from variety of numbers; texts are also stored in the same way; numbers can be grouped into bits, bytes, nibbles and words; numbers can also be described in terms of *decimal*, *binary*, *octal*, *hexadecimal* and *BCD* schemes.

Decimal Numbering

The decimal system also referred to as the *Denary* which is mostly used in everyday computational activities. It uses the digits from 0 to 9 and the number of different the base of the decimal system is 10.

The value of any digit in a number depends on its position within the number. Using 345 for instance, the 4 digit has a higher value than the digit 4 in the number 374.

For a number 7,865, it can be pronounced as *Seven thousand, eight hundred and*

sixty five. In other words, it can be written as

$$7 \times 1000$$

$$8 \times 100$$

$$6 \times 10$$

$$5 \times 1$$

these are referred to as Thousands, Hundreds, Tens and Units. We can represent this by a column as shown below.

----	10000	1000	100	10	1	1/10	1/100
		7	8	6	5		

More mathematically, the same columnar representation can be made as tabulated below.

-----	10^4	10^3	10^2	10^1	10^0	10^{-1}	10^{-2}
		7	8	6	5		

It should be noted that any number raised to the power of 0 is always equal to 1. Owing from this, it follows that any numbering system can employ column headings with base R as

	R^4	R^3	R^2	R^1	$\mathcal{P}^0$	R^1
						R^2

shown:

Binary Numbers

It is the 0 and 1 representation of numbers. Every binary number is a collection of 1's and 0's. These numbers are expressed in base 2, with the column headings looking like:

The binary number can easily be converted to its decimal equivalent. Thus, it is important that we learn this

Considering the binary number 101100-

[illegible]

16	8	4	2	1	0.5	0.25

Note that the binary number 10100 is the same as 20 in decimal.

Data lines: these are numbered from D0 upwards, while address lines are numbered from A0 onwards. So, an 8-bit wide data bus could carry numbers between 0 and 255 – a range of 256 different combinations (i.e. the bus has 8 lines giving 2 to the power 8 combinations). The number 20 on an 8-bit bus would be as in the diagram represented below.

Diagram of a shift register with 10 bits (D7 to D0). The current state is shown below the register:

Bit	D7	D6	D5	D4	D3	D2	D1	D0
Value	0	0	0	1	0	1	0	0

A bracket under the first seven bits (D7 to D1) is labeled '20', indicating a shift of 20 positions to the right.

CONVERSION

The above diagram shows how to convert a binary number into a decimal number. It is also possible to convert a decimal number into its binary equivalent. This is done by repeated division, where the decimal number is repeatedly divided by the base (in the case of binary, this is 2) until the decimal number is reduced to zero. The remainder from each division is recorded and these remainders constitute the binary number.

(Consider the case of converting the decimal number 11. Dividing by two gives five with a remainder of 1 (i.e. 11 is made up of 5 group of 2 and 1 group of 1). Dividing again by two gives:

So, $11 = 2$ group of 4, 1 group of 2 and 1 group of 1.
This is repeated thus:

So, 11 in decimal is 1011 in binary.

Note that the remainders are read upwards.

HEX NUMBERING

All data is stored and moved around in binary format no matter how the program or the user might wish to regard it or organise it. However, binary numbers can comprise very long strings of 0's and 1's when it is representing a large number. For example, the binary number 010110011001110011001 is, in fact, the decimal number 1,468,217. The binary version is hard to visualise; users find it hard to look at two binary numbers and know which is the largest. As a result, binary numbers are difficult to handle and are prone to human errors. When dealing with very large numbers, it is often converting to express the number in base 16, instead of base 10 or base 2. This is known as hexadecimal (often shortened to 'hex') and it is simple to convert from binary to hexadecimal. Also, the data on an 8-bit bus can be represented by just two alphanumeric characters instead of eight binary digits. The hex system requires symbols to represent from 0 through to 15. Since 10 to 15 are outwith normal single decimal digits, the letters A to F are used to represent 10 to 15. The table shows the relationship between binary, decimal and hex.

Binary	Decimal	Hex
0000	0	0
0001	1	1
0010	2	2
0011	3	3

0100	4	4
0101	5	
0110	6	5
0111	7	6
1000	8	7
1001	9	8
1010	10	9
1011	11	0A
1100	12	0B
1101	13	0C
1110	14	0D
		0E

Low Level Language

This is the language communicated to the computer's understanding, usually called machine language. The computer understands this language as binary (ie 0 and 1) which represents 'on' and 'off'

- *Machine Language---* Which represents instructions to the hardware in the way the machine and electronic components of the hardware is divided into:
 - *Assembly Language---* Which is the use of codes and abbreviated symbols called mnemonics to represent instructions to the computer. Usually this is enhanced by programs called translator. These are
 - ☐ Assembly-
 - ☐ Compiler-
 - ☐ Interpreter---

DOS

Disk operating system is an operating system that boots a computer to function. Any file in a Dos must be accessed via the keying of commands.

History of DOS

Before the inception of DOS, CP/M (controlled program on Microprocessor) had been in existence. It was used by special computer likes Amstrad and Apple. Later, the microsoft company came up with the development of Dos which could be installed in a fixed or hard drive of any computer unlike the cp/m that was carried about of hardware.

Operation of DOS

Command Prompt: When a computer operating under DOS is put on, alter series of primary bootings, a command prompt appears at last at the upper left of the screen. It is a field or an area where an instruction of what a computer is required to do is typed. "A" command prompt is where "A" is the default drive specification which the computer is currently working at. The drive "A" in any computer is the floppy disk

drive while "C:" is the hard disc drive (HDD) and E,C,G,F,H e.t.c. is any of the external attached drive. The prompt is followed by " : /> - ...

The *cursor* so-called is a blinking hyphen-like lighting indicating, where the next character is to be typed. As the instruction or prompt of a command is typed, and the "enter" button is pressed, the result of the instruction is displayed. The user can easily alternate between the drives to work on by typing the switch command, A:/, C:/, H:/ etc. this is dependable of the ability of the drives for effective achievement

Command: This is the instruction typed on the prompt to enable the execution of one function or the other. Example is the running of an application packages. Command are of two types basically, they are internal command and external command

INTERNAL COMMANDS- These commands remain permanently and react in the memory of the system as the computer boots. They cannot be removed or tempered. Example of these commands are

- Dir-----Directory
- Cls-----Clear screen
- Cd-----Change directory
- Md-----Make directory
- Rdk-----Remove disk
- Del-----Delete
- Copy-----
- Date-----
- Time-----

EXTERNAL COMMANDS

These commands are in the form of program files stored in the disc. It can be removed as well as tempered with. It must be in the system before it executes. Examples are

- ---erase
- ---format
- ---Find